

Aretia Protocol

A Protocol-Enforced Climate Catalyst Fund
aretiafinance.org

Abstract

Climate finance suffers from a verification problem rather than a capital problem in the narrow sense. Pledges accelerate faster than verified action, and capital that does move often does so slowly, opaquely, and without proof that it accomplished what it claimed. At the same time, permissionless token trading across major blockchain networks generates enormous, continuous transaction volume with no purpose beyond the trade itself, a volume that exists independent of any climate objective and would exist whether or not this protocol did. This paper describes a protocol that converts a fixed share of ordinary token transfer activity into standing, continuously replenished capital, termed a climate catalyst fund, without requiring any party to trust a promise at any stage. The mechanism is enforced by the token's own transfer fee extension rather than by policy, meaning the property holds by construction rather than by the good conduct of any operator; that immutability applies specifically to the aggregate fee rate, not to every parameter of the system, a distinction Section 14 makes precise. Capital leaves the fund only against independently verified project milestones, with every step, from fee withholding through final disbursement, visible on a public ledger and independently reproducible by any observer with an internet connection. This paper presents the financing and liquidity landscape motivating the design, the logic of the catalyst fund, the token protocol and its arithmetic, the project-selection and governance architecture, an economic and social analysis of the mechanism, and the impact-documentation framework applied to funded work. The result is not a pledge and not a one-time grant, but a self-replenishing source of catalytic capital engineered to keep operating whether or not anyone continues to advocate for it, conditional on the trading activity it depends upon continuing to occur.

PART I. THE FINANCING CHALLENGE FOR CLIMATE MITIGATION AND ADAPTATION PROJECTS

1. The Verification Problem in Climate Finance

Trust in climate finance breaks down at a specific and identifiable point: the gap between a pledge and its verification. Corporate climate commitments (measured as the number of companies setting near-term and net-zero targets) rose approximately 227 percent in the eighteen months leading up to mid-2025, a figure that on its own would suggest the field is accelerating.¹ Yet voluntary carbon credit retirements, the step in the lifecycle of a credit that proves it was actually used rather than purchased and left unclaimed, fell 7 percent over the same period.¹ Commitment outpaced proof through 2025, and the divergence between the two figures is itself the diagnostic.

More recent data suggests this gap is beginning to close, though from the supply side rather than the demand side. In the first half of 2026, voluntary carbon credit retirements reached approximately 98 million credits, while new issuances totaled approximately 96.8 million, a 31.9 percent decline from the 142.3 million issued in the first half of 2025.² Retirement volume held comparatively steady while issuance contracted sharply, bringing supply and demand into closer balance than in prior years. This is a healthier ratio, but it says nothing about whether verification itself has become more reliable, only that fewer new claims are being made against which verification could fail. The underlying structural weakness (that verification, where it occurs at all, happens after the fact and is frequently conducted by a party with an interest in a favorable outcome) has not been resolved by a smaller market; it has simply been tested against fewer transactions.

The mechanisms conventionally available to close the verification gap (grant programs, offset purchases, and corporate pledges among them) share a common structural weakness: each depends on a promise being honored after the money or the stated intent has already moved. A grantor rarely audits its own grantee with the same rigor an adversary would apply. A corporation announcing a pledge has every reason to publicize the announcement and comparatively little operational incentive to publicize a shortfall against it years later.

A second, related problem sits alongside the verification gap: a financing problem specifically affecting projects that have already been technically proven. A pilot project or a proven technology deployed at small scale frequently cannot secure capital to scale, not because the underlying science or engineering is in doubt, but because conventional financing pathways are built around large, discrete rounds that do not suit incremental, milestone-based scaling. A catalyst fund, capitalized continuously rather than through periodic fundraising cycles, is structurally better suited to this kind of incremental financing need than either a one-time grant or a commercial loan sized for a single large disbursement.

This is not a claim that actors in climate finance are acting in bad faith as a rule. It is a claim about incentive structure. Any system in which verification is optional, delayed, and self-administered will tend, in aggregate and over time, toward the outcome the data already shows: commitments that outpace proof. The correction implied by this diagnosis is not moral exhortation but mechanism design. If verification cannot be reliably supplied by voluntary conduct, it must be supplied by structure.

2. Why Existing Approaches Fall Short

A mechanism paper is incomplete if it does not explain what it is not. This section situates the proposed protocol against four existing categories of approach, each of which addresses part of the verification problem described in Section 1 but not, this paper argues, the whole of it.

¹ Corporate climate commitment growth and 2025 voluntary carbon credit retirement figures per the Science Based Targets initiative (SBTi), as reported in Carbon Direct, "Key Trends in the 2026 Voluntary Carbon Market" (2026).

² First-half 2026 voluntary carbon market issuance and retirement volumes per Sylvera, "Q2 2026 Carbon Data Snapshot," and Climate Focus, "Carbon Markets 2026 H1: Review and Outlook" (2026).

2.1 Voluntary Carbon Markets and the Retirement Gap

The voluntary carbon market provides a registry-based mechanism for quantifying and transacting emissions reductions, and established registries such as Verra and Gold Standard supply real methodological rigor to the question of whether a given reduction occurred. The market's weakness is not measurement but follow-through: a credit can be purchased and never retired, sitting in an account as a purchased-but-unused asset that satisfies a buyer's internal accounting without ever completing the act (retirement) that actually proves use. The retirement gap described in Section 1 is precisely this failure mode expressed as a statistic. This protocol does not replace carbon credit registries; Section 17 explicitly treats registry-issued credits as one valid form of verified output. What it adds is a disbursement mechanism, described in Section 16, that does not depend on a purchaser's subsequent discretion to complete a retirement step, since a milestone-verified grant is itself structured around confirmed completion rather than confirmed purchase.

2.2 Traditional Grant-Based Climate Philanthropy

Grant-based philanthropy, whether from private foundations, corporate social responsibility budgets, or bilateral development finance, typically disburses in large tranches against a proposal and a narrative report, with verification, if it occurs at all, taking place through a final report authored substantially by the grantee itself. This is not a claim that grant-based philanthropy is ineffective; considerable expert judgment goes into grant selection, and program officers frequently possess deep domain expertise. The structural weakness is narrower: verification and disbursement are not architecturally coupled. A grant can be fully disbursed before the deliverable is confirmed, and the confirmation, when it occurs, is rarely independent of the party being evaluated. Section 16 of this paper describes a disbursement process in which that coupling is structural rather than aspirational: a tranche is an instruction that will not execute until an independent verification event has already occurred on-chain.

2.3 Blended and Catalytic Finance

The term catalytic capital, which this paper adopts and extends in Part III, originates in blended finance practice, where it denotes capital placed specifically to unlock or de-risk activity that would not otherwise be financed on commercially reasonable terms, functioning as a first mover that makes a project bankable rather than as the entire bankroll for it. Existing catalytic capital vehicles are almost universally capitalized through discrete fundraising events: a foundation commits a sum, a development finance institution allocates a facility, and that sum, once deployed, requires a fresh fundraising cycle to replenish. The contribution proposed here is narrow but meaningful: it asks whether a catalytic capital vehicle can be capitalized continuously, as a structural property of an unrelated economic activity, rather than periodically, as the product of a fundraising campaign. Section 6 makes this precise, and Section 9 addresses directly whether the fund's activity is catalytic in the stronger, measurable sense the term implies in that literature, rather than only in intent.

2.4 Token-Based Fundraising Without Enforcement

A number of existing tokens describe themselves as supporting a charitable or environmental cause, typically through a stated intention that some share of trading fees or treasury holdings will be donated to a cause at the team's discretion. Before continuing, it is worth pausing on what a token actually is, since the term is used throughout this paper and is not universal knowledge outside blockchain contexts. A token, in this context, is a digital record of ownership maintained on a public, shared ledger called a blockchain, transferable between parties without requiring a bank, broker, or other intermediary to process the transfer. A given token's rules (how many units exist, how transfers are processed, what happens automatically at the moment of transfer) are defined in software that runs on the blockchain itself, commonly called a smart contract or, on the Solana network specifically, a program.³

³ This description of tokens and smart contracts is a general, simplified account for readers unfamiliar with blockchain terminology and is not intended as a technical or legal definition.

The distinguishing weakness of the charitable-token category, relative to the protocol proposed here, is that the donation commitment is a policy rather than a property of that underlying code. A team that states an intention to donate a share of proceeds can, without violating any smart contract, simply not do so, and an external observer has no mechanism by which to detect the omission other than the team's own subsequent disclosure. Sections 10 through 15 of this paper specify a token whose aggregate fee is a property of the mint's own on-chain configuration, verifiable by any party without trusting the issuing party's representation of it, a distinction this paper treats as load-bearing rather than cosmetic. That distinction is narrower than it may first appear, and Section 14 states precisely which parts of the fee mechanism carry this property and which do not. Several existing environmental or charitable tokens illustrate the pattern this protocol is distinguishing itself from: a project publicizes an intended percentage donation at launch, but does not implement that percentage as an enforced, on-chain transfer-fee rule, leaving compliance unverifiable after the fact.⁴

⁴ This paper does not name specific charitable or environmental tokens for comparison, since the point made is structural (policy versus code-enforced commitment) rather than an evaluation of any named project's conduct; a reader interested in specific examples can find them by searching for tokens self-described as "cause-linked" or "green" cryptocurrencies.

PART II. LIQUIDITY AVAILABILITY THROUGH EMERGING FINANCIAL INSTRUMENTS AND DIGITAL ASSETS

3. Permissionless Trading as an Untapped Capital Source

Separately from the climate finance question, permissionless token trading across major blockchain networks moves substantial real capital every day, for reasons entirely unrelated to climate objectives. This activity is not confined to any single network. On Solana, the category of permissionless, community-driven tokens reached a combined market capitalization of roughly \$40 billion with daily trading volume near \$4 billion as of mid-2026, and Solana decentralized-exchange volume alone reached approximately \$3.01 billion across a single 24-hour window.⁵ BNB Chain processed a comparable \$1.25 billion in decentralized-exchange volume over the same kind of window, with a July 2026 surge lifting BNB Chain trading volume by 45 percent and Solana on-chain activity by 38 percent, to roughly 31.4 million active addresses.⁶ On Ethereum, individual permissionless tokens in this category carry market capitalizations in the low billions of dollars (Shiba Inu at approximately \$2.5 billion and Pepe at approximately \$1.19 billion, as two illustrative examples) within a combined cross-chain category exceeding \$30 billion in aggregate market capitalization as of September 2026.⁷

This volume is not raised, solicited, or persuaded into existence on any of these networks. It is a byproduct of ordinary, permission-free markets functioning exactly as they are designed to function, and it would continue to exist in the complete absence of any climate-linked project, exactly as it did before one existed.

The observation at the center of this paper is that three facts (a verification problem in climate finance, a financing gap affecting already-proven projects, and an untapped, purpose-agnostic capital flow moving through permissionless digital-asset markets) can be connected mechanically rather than rhetorically. A protocol-level fee, enforced the same way a native transfer-fee extension is enforced on a supporting network (automatically, on every transfer, by the token standard itself rather than by the discretion of any operator), can convert a fixed share of trading activity that already exists into climate capital that does not depend on anyone continuing to believe in, advocate for, or even be aware of the underlying mission. The mechanism does not ask traders to care. It only asks the token standard to do arithmetic, which it does regardless of anyone's sentiment. This does not, on its own, establish that meaningful trading volume will in fact materialize for this specific token; Section 20 returns to that separate question directly. Section 19 returns to why Solana specifically was selected as the network on which to implement this mechanism.

4. Contributions and Organization of This Paper

This paper makes the following contributions. First, it defines the climate catalyst fund as a specific structural object, distinguished from an ordinary treasury or grant pool by two properties (continuous replenishment and non-discretionary capitalization) and situates that definition within the existing vocabulary of blended and catalytic climate finance, together with an explicit statement of the theory of change connecting the mechanism to a climate outcome. Second, it specifies the token protocol and transfer fee mechanism in full, including the arithmetic of fee allocation, the gross-up computation required to deliver a net amount through a fee-bearing internal transfer, and a complete accounting of which parts of that mechanism are enforced by immutable code and which remain subject to treasury governance. Third, it specifies a milestone-verified disbursement process, a category-specific diligence framework, and an explicit distinction between the activity, output, and outcome levels at which a funded project's progress can be described. Fourth, it presents a four-layer governance architecture designed to avoid the specific failure mode of flat, one-holder-one-vote systems once a treasury is large enough to be worth attacking, together with an account of how appointment to the technical and verification layers is kept independent of the layer it checks. Fifth, it offers a comparative economic and social analysis against network alternatives and against existing approaches to

⁵ Solana permissionless-token category market capitalization and daily volume, and Solana decentralized-exchange 24-hour volume, as of mid-2026, per CoinGecko-sourced reporting.

⁶ BNB Chain decentralized-exchange volume, and July 2026 trading-volume and active-address growth figures for BNB Chain and Solana respectively, per CryptoRank.io, "Meme Coin Trading Lifts Solana Addresses 38%, BNB Chain Volume 45%" (2026).

⁷ Ethereum-based permissionless-token market capitalizations (Shiba Inu, Pepe) and aggregate cross-chain category market capitalization, as of September 2026, per CoinGecko and CoinMarketCap-sourced reporting.

climate finance more broadly, including a direct treatment of where sustained demand for the token itself might or might not come from, on the grounds that a mechanism paper that does not explain why the alternatives were rejected, and does not explain what it does not yet know about its own token's demand, has not fully explained its own design. Sixth, it states its own risks, limitations, and open legal questions explicitly, rather than leaving them to be inferred by a skeptical reader.

The remainder of this paper is organized into five further parts. Part III defines the catalyst fund formally and states the mechanism's theory of change. Part IV specifies the token protocol: its design, its transfer-fee mechanism, the authorities that govern it, and treasury custody. Part V specifies project selection, diligence, and governance. Part VI offers an economic and social analysis, including network selection, the sources and limits of demand for the token, the mechanism's risks and limitations, and its legal and regulatory status. Part VII addresses transparency, impact documentation, and illustrative arithmetic under several volume scenarios, followed by a conclusion and an appendix of on-chain reference addresses.

PART III. THE LOGIC OF THE CATALYST FUND

5. Theory of Change

Before defining the catalyst fund formally, it is worth stating plainly the causal chain the rest of this Part and Part IV exist to make mechanically true, since a mechanism paper that specifies each link in detail can still fail to state the chain as a whole. The chain runs as follows: permissionless economic activity that already exists, and would exist regardless of this protocol, passes through a protocol-enforced fee at the moment of transfer; that fee produces continuous capital formation in a treasury-held fund, without requiring a discrete fundraising event; capital in that fund is allocated to climate projects through a selection process kept independent of the treasury itself; disbursement against a selected project proceeds only in milestone-based tranches; each milestone is confirmed by verification independent of both the protocol's own team and the project developer; verified milestones produce climate outputs, the direct, measurable products of funded work; those outputs are connected, category by category, to climate outcomes, the underlying environmental result the output is evidence of, a distinction Section 17 draws precisely; and, where a project's category and structure permit it, verified outcomes are positioned to help mobilize additional capital from other sources, the sense in which the fund aims to be catalytic rather than merely disbursing, a claim Section 9 states as a target to be measured rather than an outcome already demonstrated.

Every link in this chain up to and including milestone-based disbursement is a property of the mechanism described in this paper and is either already live on Solana mainnet or specified precisely enough to be built and independently checked. The final two links, that verified outputs correspond to genuine climate outcomes and that those outcomes in turn mobilize additional capital, depend on facts about the physical world and about other actors' subsequent behavior that no protocol, however well specified, can guarantee by construction. This paper is explicit about that boundary throughout: Section 17 separates the levels at which a project's progress can be described, Section 9 defines a measurable standard for the catalytic claim rather than asserting it, and Section 22 states directly what this mechanism does not, and cannot, guarantee.

6. Definition and Structural Properties

The catalyst fund is defined as the central object of this system: a treasury-held pool of capital that is, first, capitalized continuously by protocol-enforced transfer fees rather than by periodic fundraising; second, held under multi-party custody rather than any single controller; and third, disbursed only against independently verified project milestones rather than at the discretion of any one party. Each of these three properties is structural, meaning it is guaranteed by the design of the system rather than by the conduct of any party operating within it, a distinction elaborated further in Section 20.2.

7. Continuous Replenishment

Let $F(t)$ denote the fund balance at time t . Then:

$$F(t) = F(0) + \sum [\theta \cdot 0.02 \times V_i] \text{ for every harvested transfer } i \text{ occurring in } (0, t]$$

where V_i is the value of transfer i . $F(t)$ is non-decreasing in trading activity and requires no discrete fundraising event to increase. This is the property that distinguishes a catalyst fund, as the term is used here, from an ordinary grant pool of the kind discussed in Sections 2.2 and 2.3. A grant pool's balance is a step function, flat between fundraising events and discontinuous at them. $F(t)$ is instead a running sum over an activity, trading, that occurs continuously and

independently of the fund's own fundraising calendar, because it has none. This says nothing, however, about how large or small V_i will be in practice, or how frequently harvested transfers will occur; Section 20 treats that dependence explicitly rather than leaving $F(t)$'s good mathematical behavior to imply a guarantee about its practical scale.

8. Non-Discretionary Capitalization

No signer, founder, or committee decides whether a given transfer contributes to $F(t)$. The transfer fee extension computes and withholds the fee at the protocol level, and there is no code path in which a transfer completes without it. This property matters independently of the replenishment property described above. A fund could in principle be replenished continuously by a party who nonetheless retained discretion over whether to honor that replenishment on any given day, and such a fund would not possess the property described here. Non-discretionary capitalization means that whether a given transfer contributes to the fund cannot be revisited transaction by transaction; it does not mean that the rate at which it contributes can never be revisited at all, a distinction Section 14 addresses directly, since the aggregate fee rate itself remains subject to treasury governance rather than being fixed forever at deployment.

9. Relationship to Catalytic Capital in Blended Finance Theory

As noted in Section 2.3, the term catalytic capital is drawn from blended finance practice, where it denotes capital deployed specifically to de-risk or unlock activity that would not otherwise clear commercial financing thresholds. This protocol's catalyst fund occupies that same functional position within its own system, seeding and de-risking climate mitigation and adaptation work in verified tranches, described fully in Section 16, rather than functioning as a passive reserve awaiting a single large disbursement or as a lump-sum grant pool of the kind described in Section 2.2. The extension this paper proposes to that existing vocabulary is the substitution of a continuous, code-enforced capitalization process for the periodic, discretion-dependent capitalization process that characterizes catalytic capital vehicles as they are presently constituted.

That extension addresses how the fund is capitalized. It does not, on its own, establish that the fund's deployments are catalytic in the stronger sense the blended-finance literature typically intends: capital that demonstrably unlocks additional financing from other sources, rather than capital that is simply granted toward a climate outcome that would not otherwise have been financed at all. The two are related but distinct, and this paper's own grants, described in Part V, are catalytic in intent without yet being catalytic in demonstrated financial leverage, since no funded project has yet reached completion. To make the term measurable rather than descriptive, this paper adopts a catalytic leverage ratio, defined as:

$$\text{CLR} = (\text{Total external capital mobilized}) / (\text{Aretia capital deployed})$$

A CLR of 1 would indicate that a project received no additional financing beyond the fund's own grant; a CLR of 4 would indicate that each dollar of catalyst-fund capital was followed by four additional dollars of financing from other sources that would not otherwise have been committed. This paper does not report a CLR value, since no project has yet reached the disbursement process described in Part V against which one could be measured. It is stated here as a target metric this protocol commits to tracking and disclosing, project by project, once real disbursements occur, so that the word catalytic in this paper's own name is answerable to a number rather than resting on description alone.

PART IV. THE TOKEN PROTOCOL

The preceding parts describe why this mechanism exists and what problem it solves. This part turns to how it is actually implemented: the token itself, the fee it carries, the authorities that govern that fee, and the custody arrangement that holds what the fee collects. Readers primarily interested in the motivation and governance of the protocol, rather than its technical construction, may proceed directly to Part V without loss of continuity.

10. Token Design Overview

A token's design, in the sense used here, refers to the specific, fixed set of parameters chosen at the moment a token is created: how many units will ever exist, who if anyone retains the authority to change that later, and what if anything is automatically deducted from every transfer. These choices are made once, recorded permanently on the blockchain's own ledger, and are independently checkable by anyone, without needing to trust a description of them written by the party that created the token. Not every one of these choices is equally permanent, however, and Section 11 is explicit about which are and which are not. The remainder of this Part specifies those choices for ACT, the digital asset native to this protocol, followed by the fee mechanism the token carries and the custody arrangement governing where a withheld fee ultimately settles.

11. Supply and Authority Structure

ACT is issued as a Solana SPL Token-2022 asset. A Token-2022 mint carries several distinct on-chain authorities, each governing a different capability, and each independently revocable or retainable; treating "immutable" as a single property of the token as a whole obscures which specific capability is fixed and which is not. The following table states the current status of every authority relevant to this protocol, and is intended to be read in full rather than summarized as a single claim.

Authority	Current holder	Can modify?	Revoked?
Mint authority	None	No	Yes, permanently, following the single initial mint
Freeze authority	None	No	Yes, never granted at deployment
Transfer-fee-config authority	Treasury multisig (2-of-3 Squads)	Yes, with 2-of-3 signer approval	No
Withdraw-withheld authority	Treasury multisig (2-of-3 Squads)	Yes, with 2-of-3 signer approval	No
Metadata authority	None; no metadata extension exists on this mint	No	Not applicable; the capability was never reserved and cannot be added retroactively
Treasury (vault) authority	Treasury multisig (2-of-3 Squads)	Yes, with 2-of-3 signer approval	No, and not intended to be; this authority is meant to remain active and exercised

Two properties are fixed absolutely, by permanent revocation, and cannot be reinstated by any party under any circumstance: the total supply, at 100,000,000 units with 9 decimals, and the impossibility of freezing any individual account holding ACT. Mint authority was permanently revoked immediately following the single initial mint, which deposited the full supply into the treasury vault's associated token account, meaning the ability to create additional

ACT was destroyed, not merely declined to be used. No wallet, including the treasury's own, has held newly created ACT outside that initial mint. Freeze authority was never granted at deployment, meaning no account holding ACT can be frozen by any party, including this protocol's own operating entity.

Two further authorities are, by contrast, deliberately retained by the treasury multisig rather than revoked, and this paper states plainly what that means: the transfer-fee-config authority, which can change the fee rate itself, and the withdraw-withheld authority, which can move harvested fees out of the accounts where the withholding mechanism accumulates them. Both require the same 2-of-3 multisig approval that governs every other treasury action described in Section 15; neither can be exercised by a single signer, including the founder, acting alone. This is not a theoretical possibility: the fee rate specified elsewhere in this paper, 3.5 percent, was arrived at through exactly this mechanism, changed from an earlier 4.1 percent by a 2-of-3 approved transaction, with Token-2022's own protocol rules requiring that the change take effect only from a specified future epoch rather than immediately or retroactively. Section 14 states precisely what this means for which parts of the fee mechanism can and cannot be characterized as immutable. On-chain addresses for the mint and the accounts referenced throughout this Part are collected in Appendix A, rather than repeated inline.

12. Immutability as a Design Choice, and Its Costs

The irrevocability described in Section 11 is a deliberate tradeoff rather than a costless guarantee, and this section states its known costs directly rather than treating them as incidental. Revoking mint authority forecloses, permanently, any future ability to attach on-chain identifying metadata (a name, symbol, or logo) to the mint, since the standard mechanisms for doing so, Token-2022's own metadata extension and the separate Metaplex Token Metadata program, each require an authority this mint no longer possesses or a metadata slot that was never reserved in the account's byte layout at creation. The practical consequence is a known protocol trade-off rather than a minor cosmetic issue: ACT displays without a name, symbol, or logo in wallets, block explorers, and automated token-risk scanners, several of which flag the token using heuristics designed to detect an active rug pull rather than the actual condition present, a fixed and immutable supply with no metadata. This can cause genuine user confusion, wallet-display and exchange-integration friction, and a real risk that a fraudulent copy of ACT, minted separately under a similar-looking name, could be mistaken for the genuine asset precisely because the genuine asset displays no distinguishing identity of its own. This is treated here as an honest cost of the commitment device rather than an oversight: because the token's own identity cannot be made self-evident on-chain, the mitigation adopted is to make independent verification of its actual, real properties as easy as possible for anyone who encounters it, and to direct anyone evaluating an unfamiliar or unverified-looking token claiming to be ACT to that verification tool, described in Section 24.3, rather than to any on-chain name or logo, which this token will never display.

13. Market Access and Permissionless Trading

ACT trades on the open market against USDT, with no permission requirement and no gatekeeping of any kind. Any Solana wallet or exchange capable of handling a standard Token-2022 asset can transact it without integrating any protocol-specific logic, a property inherited directly from the token standard rather than engineered specifically for this project.

14. The Transfer-Fee Mechanism

14.1 The Withholding Function

Token-2022's native transfer fee extension withholds a percentage of every transfer at the protocol level, before the recipient's balance is credited, up to a configured maximum absolute amount so that a single very large transfer does

not withhold an unbounded sum. The fee rate is 350 basis points, or 3.5 percent of the transfer amount, as of the epoch at which this rate last took effect.

14.2 What Is Protocol-Enforced and What Is Treasury Policy

This distinction is stated once here, prominently, because it governs how every other claim in this paper about the fee mechanism should be read. What Token-2022 enforces, at the protocol level, on every transfer, without exception and without requiring any party's cooperation, is that the aggregate withheld amount is computed and withheld correctly according to whatever rate is currently configured on the mint. What Token-2022 does not enforce, because the token standard has no native concept of it, is any particular destination or split for that withheld amount once harvested, and it does not enforce that the rate itself stays fixed forever, since the transfer-fee-config authority described in Section 11 retains the power to change it, subject only to 2-of-3 multisig approval and Token-2022's mandatory future-epoch delay.

Concretely: the correct way to state this protocol's strongest technical claim is that the protocol mechanically withholds 3.5 percent of every ACT transfer, and that the treasury subsequently routes harvested fees according to the published 2.0 percent, 1.0 percent, and 0.5 percent allocation framework described in Section 14.3. It is not accurate to state that the protocol automatically sends 2.0 percent of every transfer to climate projects, since the routing of harvested fees to their destinations, described in Section 14.4, and the allocation split itself, are treasury procedure carried out under multisig approval, not a second on-chain-enforced ratio. This is a narrower claim than the aggregate withholding claim, and this paper treats the narrower, accurate claim as the one worth making.

14.3 Fee Allocation

The withheld fee divides as follows: 96.5 percent of every transfer reaches the recipient net of the fee; of the 3.5 percent withheld, 2.0 percentage points (approximately 57.1 percent of the withheld amount) are routed to the catalyst fund, 1.0 percentage point (approximately 28.6 percent of the withheld amount) to liquidity, and 0.5 percentage points (approximately 14.3 percent of the withheld amount) to protocol management, once harvested. As Section 14.2 states, only the 3.5 percent aggregate is enforced by the mint's on-chain configuration; the destination split above is the treasury's own harvest-and-route procedure, applied consistently every time withheld fees are harvested, under the multisig custody described in Section 15.

14.4 The Two-Phase Withhold-and-Harvest Design

A withheld amount does not move to its destination automatically at transfer time. Instead, the mechanism operates in two phases. In the first phase, at the moment of every transfer, the withheld amount accumulates inside the recipient's own token account as a running balance, publicly readable by any observer. In the second phase, an authorized party periodically executes a separate "harvest" instruction that sweeps accumulated withheld amounts, across every account where they have built up, into the treasury. This two-phase design (withhold locally at the moment of transfer, then harvest periodically in a separate step) is what allows every wallet, exchange, and routing aggregator already supporting Token-2022 to handle ACT correctly with no protocol-specific integration whatsoever. The complexity of routing the fee to its final destination is confined entirely to the harvest step, which only the treasury multisig executes (that is, which only takes effect once a sufficient number of the treasury's independent, designated signers have separately approved it, per the custody model described in Section 15) rather than being distributed across every individual transfer.

14.5 Management-Fee Delivery and the Gross-Up Problem

The management fee wallet is held separately from the treasury multisig (see Appendix A for its address), so that operational compensation to the protocol's operating entity can never be commingled with catalyst fund capital structurally, rather than merely by reporting convention. Delivering that compensation requires a second transfer, from

the treasury vault to the management wallet, and that second transfer itself incurs the same withholding described in Section 14.1, since the fee applies to every transfer of the token without exception. To land a net amount N in the management wallet after that second transfer's own fee is deducted, the treasury must send a gross amount G satisfying:

$$\begin{aligned} G - \text{fee}(G) &= N \\ G \times (1 - b) &= N \text{ [when } \text{fee}(G) < M] \\ G &= N / (1 - b) = N / 0.965 \end{aligned}$$

where b is the fee rate defined in Section 14.1 and M is the mint's configured maximum fee. For example, to deliver a net N of 1,000 ACT to the management wallet, the treasury must send a gross G of 1,000 divided by 0.965, approximately 1,036.27 ACT, of which approximately 36.27 ACT is withheld on that second transfer and re-enters the fee-split pool described in Section 14.3. This recursive fee-on-internal-transfer problem does not appear to have been treated explicitly in prior public documentation of comparable mechanisms, and is included here because a treasury operator who failed to account for it would systematically underdeliver the intended management compensation on every occasion the calculation was performed.

15. Treasury Custody

15.1 Multisignature Custody Model

Harvested fees settle into a Solana multisig, using the Squads protocol, live on mainnet (see Appendix A for its address). The multisig defines a signer set of three designated parties and a threshold of two: a transaction executes only once at least two of the three have independently approved it. No single key, including any key held by this protocol's operating entity or by any individual associated with it, can move catalyst fund capital unilaterally, and no single key can unilaterally change the transfer-fee rate described in Section 11 either, since that authority is held by the same multisig under the same threshold.

15.2 Threshold Selection: Why Two of Three

A threshold of two of three was selected as the smallest configuration that eliminates unilateral control while remaining operationally practical for an entity of this size, consistent with current guidance from Squads' own documentation, which recommends configurations such as two-of-three or three-of-five over a single signer or a threshold equal to the full signer set. A larger signer set with a proportionally larger threshold would reduce the marginal influence of any single compromised or unavailable signer further, at the cost of requiring coordination among a larger group for every routine transaction. Two of three is treated here as an appropriate starting point given the treasury's current scale; Section 18's governance architecture provides the mechanism by which this parameter, like others, could be revisited as the system matures.

15.3 Signer Identity and Independence

[Placeholder: this subsection will state, for each of the three signers, their role and relationship to the protocol, their independence from one another (including whether any two signers are economically related), their jurisdiction, the conflict-of-interest policy governing their participation, the procedure for replacing a signer who becomes unavailable or is removed, and the key-security practices each signer follows. Institutional counterparties evaluating this treasury's custody arrangement will expect this information in full, not only the wallet addresses collected in Appendix A, and this section is left open pending that disclosure rather than answered with generic assurances.]

15.4 Operational Verification

The propose, review, approve, and execute cycle has been tested end to end with a real transaction on mainnet, approved independently by two of the three signers, rather than verified only in principle: the transfer-fee rate change described in Section 11 is itself an example of this cycle exercised for real, not merely configured. The distinction between a custody model that has been exercised and one that has merely been configured is treated here as material enough to state explicitly.

PART V. PROJECT SELECTION AND GOVERNANCE

16. Milestone-Verified Disbursement

16.1 Forms of Disbursement

Capital leaves the catalyst fund primarily as a grant to a vetted climate project partner not itself tokenized, such as a reforestation cooperative or a community solar developer. A second form (an on-chain retirement of tokenized carbon credits sourced from an established registry, which would produce a quantified tonnes-of-carbon-dioxide-equivalent figure tied to a specific transaction hash) is not implemented in the current design and is not available as a disbursement path today. It is recorded here as a form the protocol may adopt in the future, should tokenized-credit infrastructure and registry practice develop to a point that makes it a workable second approach, rather than as a mechanism presently in use alongside direct project grants.

16.2 Tranche Structure and the Sequencing Problem

Disbursement does not release as a single lump sum against an initial approval. An approved project's total commitment is instead divided into a series of tranches, with each tranche releasing only after independent verification confirms that the preceding milestone was actually met. This sequencing is the structural mechanism that prevents the statement "the fund financed a project" from silently coming to mean, in a given instance, "the fund sent money to a project that subsequently failed to complete its work," a failure mode this paper argued in Section 2.2 is endemic to disbursement processes in which verification and payment are not architecturally coupled.

16.3 On-Chain Auditability of Disbursements

Every disbursement is a public on-chain transaction originating from the known treasury address (see Appendix A). No disbursement is representable, to any observer, as anything other than what the underlying chain state shows it to be, a property that holds regardless of what any party subsequently claims about it. The criteria used to determine which projects reach this disbursement process in the first place are given in Section 17.

17. Project Selection and Diligence

17.1 Category Taxonomy

The catalyst fund finances five categories of climate work: renewable energy, waste-to-energy management, carbon reduction, technical assistance, and flood management and reforestation. Each category admits a structurally different kind of verifiable claim, and each is accordingly held to a diligence standard appropriate to that specific claim, applied by the technical committee described in Section 18 before a project is admitted to the funding slate.

17.2 Levels of Verification: Activity, Output, and Outcome

A category-specific diligence standard, stated on its own, can obscure an important distinction this paper draws explicitly: verifying that a claimed result occurred is not the same as verifying that the claimed result is itself the climate outcome the fund exists to produce. Three levels are distinguished throughout this Part. The activity level describes what work was performed: a solar array was installed, a reforestation planting campaign was carried out. The output level describes what was directly produced by that activity, and is typically what a category's diligence standard in Section 17.3 verifies: metered generation capacity added, hectares planted. The outcome level describes the underlying climate result the output is evidence of, which frequently requires an additional inferential step beyond the output measurement itself: metered generation capacity translates to avoided emissions only via an assumption

about what energy source it displaces; hectares planted translates to tonnes of carbon dioxide equivalent removed only via an assumption, typically registry-governed, about growth rates, survival rates, and measurement methodology over time. This protocol's disbursement process, described in Section 16, verifies milestones at the activity and output levels directly, since those are the levels a project's own progress reports and independent field verification can confirm within the timeframe of a funding tranche. Outcome-level claims, where reported, rely on established third-party methodologies, such as registry protocols for carbon accounting, rather than on a claim this protocol makes independently, a distinction this paper considers necessary to state given how easily output and outcome are conflated in climate-finance reporting generally.

17.3 Category-Specific Diligence Standards

Category	Verified output	Diligence standard
Renewable energy	Metered generation capacity added	Independent engineering certification of installed and operating capacity; interconnection or off-take documentation
Waste-to-energy management	Waste diverted and energy generated	Facility operating permits, throughput metering, emissions monitored against a stated baseline
Carbon reduction	Emissions avoided or removed	Registry-issued credits, from Verra, Gold Standard, or a comparable standard, or an equivalent third-party MRV protocol
Technical assistance	Capacity delivered: studies, training, MRV systems built	Deliverable-based milestones; independent review of the technical work product itself, since there is no physical asset to inspect
Flood management and reforestation	Hectares restored; flood-risk reduction achieved	Remote-sensing canopy verification for reforestation; hydrological modeling or post-event impact assessment for flood management

Each row verifies an output in the sense defined in Section 17.2. Translating a given row's output into a quantified outcome, such as tonnes of carbon dioxide equivalent avoided, depends on the category-appropriate third-party methodology referenced in that row, and is reported as such rather than presented as a claim this protocol verifies independently.

17.4 The Special Case of Technical Assistance

Technical assistance differs from the other four categories in kind rather than merely in degree. Where renewable energy, waste-to-energy, carbon reduction, and flood management and reforestation projects each produce a physical or registry-verifiable output that exists independently of any narrative account of it, technical assistance instead funds the capacity that makes a future project bankable in the first place: a feasibility study, an MRV system a smaller developer could not otherwise afford to build on its own, or training delivered to a local partner's own monitoring staff. Its diligence standard is accordingly oriented around the work product actually delivered against a defined scope of engagement, rather than a physical inspection of an asset in the field, since in this category no such asset yet exists to inspect.

17.5 Portfolio Weighting as a Governance Parameter

A project's category determines which diligence standard from Section 17.3 applies to it. It does not alter the milestone-verified release mechanism defined in Section 16, which governs every category identically regardless of

the nature of the underlying claim. The relative weighting of categories within the overall portfolio is a Layer 1 governance parameter under Section 18, revised through the same procedure that sets any other funding priority, and is accordingly not fixed by this document.

18. Governance

18.1 The Failure Mode of Flat Token-Holder Voting

A flat, one-holder-one-vote system governing individual disbursements fails once a treasury is large enough to be worth attacking, because a well-organized proposal can extract funds from a naively administered vote regardless of that vote's formal legitimacy. This is not a hypothetical concern specific to this protocol; it is a general property of governance systems in which the entire decision authority over a valuable resource is concentrated in a single voting mechanism with no independent check on the substance of what is being voted upon.

18.2 A Four-Layer Architecture

Governance under this protocol is layered instead of flat, across four distinct functions. The first layer is community governance, exercised by token holders through Realms, which sets funding priorities, project categories, and treasury allocation weights, but does not vote on individual disbursements. The second layer is a technical committee, which screens individual project applications for climate-science credibility, MRV soundness, and financial feasibility before a project can be admitted to the process described in Section 16. The third layer is independent verification, which confirms, at each milestone, that a party independent of both the protocol's own team and the project developer has verified the claimed progress actually occurred. The fourth layer is smart-contract execution, which releases funds only once the second and third layers have both signed off on a given milestone, with no human discretion exercised at the point of transfer itself.

18.3 Appointment Independence

Layering discretion across four functions addresses concentration within a single vote, but it introduces a narrower version of the same problem one level removed if left unaddressed: if the community layer alone appoints the technical committee, and the technical committee alone appoints the independent verifier, the two checking layers can in principle become answerable only to each other rather than genuinely independent of the process they check. This protocol accordingly separates the appointment path from the layers it appoints. The community layer sets strategic priorities rather than directly selecting individuals for the technical committee; appointment to the technical committee, and separately to the pool of independent verifiers, runs through a nomination process kept structurally distinct from both the community vote and the treasury's own operating entity, so that neither the committee nor the verifier owes its position to the party whose work it is evaluating. The technical committee approves the methodology by which a milestone is assessed; the independent verifier confirms that a specific milestone was met under that methodology; the treasury executes only once both have signed off. This sequencing, appointment kept separate from evaluation kept separate from execution, is the specific mechanism by which this protocol avoids recreating, one layer later, the centralization problem the four-layer architecture in Section 18.2 was built to solve at the level of individual disbursements.

18.4 Phased Activation

Governance is phased in against real participation rather than switched on in full at launch. Layer 1 activates once the holder base is large enough for a vote to carry genuine information content rather than being symbolic in practice. Layers 2 through 4 build out across subsequent phases of the protocol's roadmap. Until Layer 1 is live, the treasury operates on team-curated grants under the identical milestone-verification discipline described in Section 16, meaning the sequencing guarantee described there does not depend on governance being fully activated to hold.

18.5 Comparison to Single-Signer and DAO-Only Models

It is worth situating this architecture against the two more common alternatives. A single-signer model, in which one founder or entity retains full discretion over disbursements, offers speed and simplicity at the cost of the exact non-discretionary property this paper has emphasized throughout; it is a system whose good behavior depends entirely on the continued good faith of one party. A DAO-only model, in which token holders vote directly on every disbursement with no intervening technical or verification layer, addresses the concentration-of-discretion problem but reintroduces the failure mode described in Section 18.1: a majority or a well-coordinated plurality of token holders can, in principle, vote to disburse the treasury to itself. The four-layer architecture described in Section 18.2, together with the appointment independence described in Section 18.3, is designed to occupy neither extreme, distributing discretion across community priority-setting, technical screening, independent verification, and mechanical execution such that no single layer, including a hypothetical majority of token holders, can unilaterally move funds or appoint the parties who check its own decisions.

PART VI. ECONOMIC AND SOCIAL ANALYSIS

19. Network Selection

19.1 Requirements for the Underlying Ledger

The mechanism described in this paper requires a ledger with three properties: transaction costs low enough that routine harvest and disbursement operations remain economical at the scale this treasury is likely to operate at; a native, protocol-level fee-on-transfer primitive, so that the withholding described in Section 14.1 is a property of the token standard rather than of custom program logic that would otherwise need to be written and maintained; and sufficient liquidity in USDT or an equivalent stable asset to support the trading pair described in Section 13.

19.2 Comparative Analysis

Network	Fee per transfer	Native fee-on-transfer support	USDT liquidity
Solana	approximately \$0.0003	Yes, via the Token-2022 extension	Native SPL USDT, deep on Raydium and Orca
Base	approximately \$0.01 to \$0.05	No, requires custom contract logic	Bridged USDT
BNB Chain	approximately \$0.10 to \$0.30	No, requires custom contract logic	Deep, but with more centralized custody risk
Ethereum L1	\$2 to \$30 or more	No, requires custom contract logic	Deepest overall, but expensive to disburse from frequently

19.3 The Cost of Bespoke Fee Logic

The deciding factor in Section 19.2 is the third column, native fee-on-transfer support, rather than the first. On every chain lacking a native primitive of this kind, the withholding and routing mechanism specified in Part IV would have to be implemented as bespoke program logic written and maintained specifically for this project, introducing an additional attack surface with its own potential defects, one that would need to be reviewed separately from, and in addition to, the underlying token standard itself. Token-2022's transfer fee extension is instead a property of shared infrastructure that every Solana wallet, exchange, and routing aggregator already implements correctly, having been built once and used by the entire ecosystem rather than once per project. Building the equivalent mechanism on a chain without this primitive would mean asking every counterparty to trust this protocol's own custom code for a function the token standard otherwise guarantees at the infrastructure level.

20. Economic Analysis

20.1 Fee Incidence and Trading Behavior

A transfer fee of 350 basis points is a real cost borne by anyone transacting the token, and it is fair to ask whether a fee of this magnitude meaningfully deters the trading activity the mechanism depends upon in the first place. This paper does not claim to have resolved that question empirically, since no ACT trading history yet exists to measure against, and it treats the fee's effect on trading behavior as an open empirical question rather than as one this document can settle by comparison to other tokens. What can be stated without empirical trading data is narrower: the fee is fixed and fully disclosed prior to any transaction, which distinguishes it from slippage or price impact, both of which vary

with trade size and pool depth and are frequently larger in absolute terms for a given trade than the fee itself, and a rational trader prices a known, fixed cost into their decision differently than an uncertain, variable one. Once real trading activity exists, this protocol commits to measuring, and disclosing, the fee's actual effect through a specific set of metrics: trading volume over time, the number of unique holders, transfer frequency, average transaction size, realized fee revenue, available liquidity, typical slippage, and holder retention. Treating 3.5 percent as a design parameter whose effect remains to be measured, rather than as a figure already vindicated by comparison to other assets, is the more defensible framing this paper adopts.

20.2 Credible Commitment Through Mechanism Design

The distinction drawn in Section 2.4, between a policy commitment and a code-enforced property, has a formal name in economics: the difference between cheap talk and a credible commitment device. A statement of intent that the issuer could costlessly reverse without technical consequence is cheap talk, regardless of the sincerity with which it is made. A property enforced by an immutable, permissionless token standard, verifiable by any party without trusting the issuer's account of it, is a credible commitment in the technical sense: the cost of reneging is not merely reputational but architectural, since reneging on that specific property would require an authority that no longer exists. Sections 11 and 12 describe the specific mechanism, permanent revocation of mint and freeze authority, by which this commitment is made costly to break for supply and account freezing specifically, and the cost that this same irrevocability imposes elsewhere; Section 14.2 is explicit that this credible-commitment property applies to the aggregate fee rate's correct withholding, not to the rate's value remaining fixed forever or to the destination split of harvested fees, both of which remain governed rather than immutable.

20.3 Liquidity Provision Incentives and Impermanent Loss

A separate incentive question concerns the party or parties who eventually supply the other side of the ACT-USDT trading pair described in Section 13, since a liquidity provider to an automated market maker pool bears a distinct risk, commonly termed impermanent loss, in which a divergence between the pool's two assets can leave a liquidity position worth less, in dollar terms, than simply holding the two assets separately outside the pool. This risk is a standard, well-documented property of automated market making rather than a feature specific to this token. It is noted here because an economic analysis of the fee mechanism that omitted the liquidity side of the market would be incomplete: the catalyst fund's continuous replenishment, described in Section 7, depends on trading volume, and trading volume depends on liquidity existing in the first place, which in turn depends on some party accepting impermanent loss risk in exchange for a share of trading fees.

20.4 Sources of Demand for ACT

This mechanism explains how trading activity generates climate capital. It does not, on its own, explain why sustained trading activity in ACT specifically should occur, and this paper is explicit that these are two separate questions rather than one. The first question, whether trading activity can generate climate capital, is answered affirmatively and mechanically by Sections 7 and 8: the arithmetic holds regardless of who is trading or why. The second question, whether ACT can sustain sufficient economic activity for that capital generation to become material at scale, is unresolved and cannot be resolved by this document; it depends on market conditions this protocol does not control. This paper's own conclusion already acknowledges that the primary remaining challenge is scale and real trading volume rather than mechanism design, and that acknowledgment is stated here as a load-bearing qualification of every figure in Section 25, not as a closing caveat to be read past.

It is worth separating, further, what ACT actually is from what its holders may independently believe about it. Five distinct things are commonly conflated under the label "why buy the token," and this paper treats them separately. Protocol utility is what ACT does inside this system: it is the asset whose transfers trigger the fee described in Part IV. The economic mechanism is what happens when ACT is transferred: a fixed percentage is withheld and eventually

routed as described in Section 14. Governance rights are what holders eventually control, once Layer 1 described in Section 18.4 activates: funding priorities and category weights, not individual disbursements. Impact utility is what the resulting activity finances: verified climate work, described in Part V. Speculative demand is whatever the market independently creates around the token's price, a phenomenon this protocol neither engineers nor controls. This paper does not claim that climate impact automatically creates token value, and does not require that claim for its mechanism to function: the fee-generation arithmetic in Section 7 operates on transfer activity itself, not on price appreciation, and would produce identical catalyst-fund inflows whether ACT's price rose, fell, or stayed flat, provided the same volume of transfers occurred. The protocol does not require a claim that ACT will appreciate in value for its stated purpose to be met.

21. Social Dimensions of Project Selection

An economic analysis of this mechanism is incomplete without a corresponding account of who it is designed to reach. The five-category taxonomy in Section 17.1 was chosen deliberately to include technical assistance, described in Section 17.4, precisely because that category serves a population conventional project financing tends to overlook: developers with a credible technical approach but insufficient resources to produce the feasibility studies, MRV systems, or training infrastructure that would make a subsequent capital request bankable elsewhere. A diligence framework calibrated only to well-resourced applicants (one requiring extensive documentation, formal certifications, and administrative capacity as a precondition of consideration) would systematically exclude exactly the applicants a catalyst fund of this kind is best positioned to serve. The category-specific diligence standards in Section 17.3 are accordingly scoped to what each category can actually produce as evidence, rather than to a single uniform documentation burden applied regardless of an applicant's size or resources. This is stated as a design intention rather than a guarantee; whether the diligence process in practice remains accessible to smaller and less-resourced developers, as opposed to only the applicants most able to navigate a formal review process, is a question this paper does not consider settled and returns to as an open item for future evaluation.

22. Risks and Limitations

This paper has argued for the mechanism it describes throughout; this section states plainly what could cause it to fail or to matter less than intended, rather than leaving those conclusions to be inferred. Insufficient trading volume is the most direct risk: the fund's inflow is a fixed percentage of transfer activity, and if that activity remains small, the fund remains small regardless of how sound the underlying mechanism is, a dependency Section 20.4 states explicitly. A fee-induced reduction in trading activity is a related but distinct risk: it is possible that the 3.5 percent fee itself measurably suppresses the volume it depends upon, a question Section 20.1 commits to measuring rather than assuming an answer to in either direction. Liquidity risk and token price volatility are ordinary properties of any freely traded digital asset and apply to ACT without qualification; a thin or newly seeded liquidity pool is particularly exposed to large price movements from comparatively small trades. Smart-contract and token-standard risk is present at the level of the underlying infrastructure: Token-2022 is a Solana Labs–maintained standard used broadly across the ecosystem, but no software is free of the possibility of an undiscovered defect, and this protocol's own custom code, including the harvest-and-route procedure described in Section 14.4, has not yet undergone the independent program review noted as outstanding in Section 26.

Governance capture is a risk this paper's four-layer architecture, described in Section 18, is designed to resist rather than a risk it claims to have eliminated; a sufficiently well-resourced or coordinated actor could still attempt to influence the community layer, the nomination process described in Section 18.3, or the composition of the technical committee or verifier pool over time, and no governance design is permanently immune to a sufficiently determined and well-funded attempt at capture. Verifier conflicts are a related risk specific to the independent-verification layer: a verifier who is nominally independent but has an undisclosed relationship to a project developer or to the protocol's

own operating entity would undermine the sequencing guarantee described in Section 16.2 without that failure being visible from the on-chain record alone, since the chain records that a verification event occurred, not the private relationships of the party who performed it. Project failure is a risk inherent to project financing generally: a funded project may fail to complete its work even where the fund's own disbursement discipline functioned exactly as designed, and the tranche structure described in Section 16.2 limits the fund's exposure to a failed project but does not eliminate it. Regulatory uncertainty, addressed further in Section 23, is a risk that applies to the protocol's own legal characterization and could constrain who may participate in it, how it may be marketed, or how it may operate in a given jurisdiction, in ways this paper cannot predict or foreclose. Climate-impact measurement uncertainty, addressed in Section 17.2, means that even a fully successful disbursement process, executed exactly as designed, produces verified outputs whose translation into verified climate outcomes still depends on third-party methodologies this protocol does not control and cannot independently guarantee.

None of these risks is hypothetical in the sense of being unlikely; several, including insufficient trading volume and the absence of a completed independent program review, are simply true of the mechanism's current, early stage rather than uncertain possibilities. They are stated here together, rather than distributed only as caveats within the sections they most directly concern, so that a reader assessing this mechanism does not have to reconstruct its risk profile from scattered qualifications.

23. Legal and Regulatory Status

This paper deliberately does not attempt a legal classification of ACT, since doing so credibly requires jurisdiction-specific analysis this document is not positioned to provide, and a technical architecture, however carefully specified, does not by itself settle a legal question. ACT is a blockchain-based digital asset native to the Aretia Protocol. Nothing in this document constitutes an offer of securities, investment advice, or a representation that ACT is classified identically across jurisdictions, whether as a utility token, a payment token, an investment contract, a governance token, a commodity-like digital asset, a financial instrument, a collective investment arrangement, or otherwise. The legal and regulatory treatment of ACT may differ by jurisdiction and depends on the specific circumstances of its issuance, distribution, trading, and use, including circumstances that have not yet occurred as of this document's publication. Prospective holders, and any party considering a transaction involving ACT outside the ordinary course of open-market trading described in Section 13, should seek jurisdiction-specific legal advice rather than rely on this document's technical description of the protocol as a substitute for that advice.

PART VII. IMPACT DOCUMENTATION AND MANAGEMENT

24. Transparency and Independent Verifiability

24.1 On-Chain and Off-Chain Verification

This paper distinguishes two categories of claim rather than treating all of its claims as equally verifiable by the same means. On-chain claims, including the mint's fixed supply and authority structure described in Section 11, the transfer fee configuration described in Section 14.3, the treasury's multisig threshold and signer set described in Section 15.1, and every disbursement described in Section 16.3, are descriptions of chain state that already exists and can be independently recomputed by any party from the Solana ledger itself, without trusting this document's account of them. Off-chain claims, including the climate statistics cited in Part I, engineering and MRV certifications referenced in Section 17.3, and the real-world project performance, social outcomes, and physical climate results a funded project reports, cannot be verified from the Solana ledger, because the ledger records that the protocol did what it claims to have done, such as executing a disbursement, and not independently whether the underlying real-world claim the disbursement was made against, such as a stated quantity of avoided emissions, is itself true. The ledger verifies protocol behavior; it does not verify physical reality. This protocol's architecture is designed so that what the ledger can verify, it verifies completely and without requiring trust in this document's own representation; what the ledger cannot verify is instead routed through the independent, off-chain verification layer described in Section 18.2, and this paper is explicit that the two are different kinds of assurance rather than a single undifferentiated claim of verifiability.

24.2 The Transparency Dashboard

A public transparency dashboard, currently in development and not yet live, is intended to index this activity as a running ledger (inflow, disbursement, recipient, and running total), presented continuously rather than as a periodic report published when convenient for the operator. It is not yet complete; it is noted here only to distinguish it from the mechanism already described in Section 24.3, which is complete and live today.

24.3 Client-Side Self-Verification

In addition to the dashboard described above, a client-side verification tool has already been deployed at this protocol's public website. This tool queries Solana mainnet directly from the visitor's own browser, with no intervening server in the request path at any point, and independently recomputes each of the on-chain claims made in Sections 11 and 14.3 (namely the mint's authorities, its supply and decimal configuration, the transfer fee rate, the fee authorities, and the treasury's token balance), displaying the live, on-chain result beside the documented claim for direct comparison. The tool's own source is visible in the page from which it runs, so that a skeptical visitor is not asked to trust the verification tool either. This is treated here as a partial but genuine answer to the immutability cost described in Section 12: because the token cannot display a verified identity on-chain, the mitigation instead makes the underlying on-chain facts trivially checkable by anyone who arrives skeptical of them, though it verifies only the on-chain category of claim described in Section 24.1, not the off-chain category.

25. Illustrative Calculation

The mechanism's output is arithmetic, not a forecast, and this section presents that arithmetic under several illustrative scenarios, chosen for round numbers rather than as a projection of what volume will actually occur. Every figure in this section should be read together with Section 20.4: nothing below states or implies that any particular row will resemble actual future volume. For any aggregate ACT transfer volume V occurring between harvests, and before the maximum-fee cap described in Section 14.1 applies to any individual transfer within that volume, the following table

applies the fee-split proportions from Section 14.3 at three illustrative per-harvest scales.

Aggregate volume V	Catalyst fund (2.0%)	Liquidity (1.0%)	Management fee (0.5%)
\$10,000	\$200	\$100	\$50
\$100,000	\$2,000	\$1,000	\$500
\$1,000,000	\$20,000	\$10,000	\$5,000

The same arithmetic, applied to illustrative annual transfer-volume scenarios rather than a single harvest, shows the scale economics of the mechanism more directly:

Illustrative annual transfer volume	Illustrative annual catalyst-fund capital
\$1,000,000	\$20,000
\$10,000,000	\$200,000
\$50,000,000	\$1,000,000
\$100,000,000	\$2,000,000
\$500,000,000	\$10,000,000
\$1,000,000,000	\$20,000,000

Both tables are illustrative scenarios, not forecasts, and neither should be read as a projection of ACT's actual future trading volume, which is presently unknown. No step in either computation depends on price, sentiment, or continued attention to the protocol's mission; each is a fixed function of transfer volume, computed identically whether that volume reflects one large trade or many small ones. Actual accrual to the catalyst fund is a direct, mechanical function of real trading volume, which this paper does not forecast anywhere.

26. Conclusion

This paper has described a mechanism that converts a fixed, protocol-enforced share of ordinary token-transfer activity into standing capital for climate mitigation and adaptation work, a catalyst fund that requires no renewed pledge to keep functioning, no trust in any single party's discretion to keep it honest, and no step, from fee withholding through milestone-verified disbursement, that cannot be independently checked against the public ledger by any party willing to do so. The mechanism, as distinct from its eventual scale, is live: ACT is deployed on Solana mainnet with its supply fixed and mint authority revoked, and the treasury multisig operates under tested, exercised, multi-party custody rather than merely configured custody. This paper has also been explicit about what remains outside the mechanism's own guarantees: the fee rate and the destination split of harvested fees remain governed rather than immutable, as Section 14.2 states; sustained demand for ACT itself is an open question Section 20.4 does not resolve; the legal characterization of ACT varies by jurisdiction and is not settled by this document, as Section 23 states; and the risks enumerated in Section 22 are properties of the mechanism's current stage, not remote possibilities.

What remains, beyond those open questions, is scale: real trading volume, a seeded liquidity pool, an independent program review, a live transparency dashboard, disclosed signer identities under Section 15.3, and a first slate of verified project partners against which the catalytic leverage ratio defined in Section 9 can finally be measured rather than only defined. None of these remaining items requires trusting a new promise to complete; each is a specific, named piece of work whose completion will itself be as verifiable, once done, as everything described in the sections above already is. The proposition at the center of this document is narrower than a claim that this mechanism will succeed at scale, which no document can guarantee in advance: this protocol does not ask any party to promise that climate finance will happen. It attempts to encode the financing mechanism into the infrastructure that generates the capital, while keeping capital generation, project selection, verification, and disbursement structurally separate from one another, and while stating, as precisely as this document is able to, exactly where that architecture's guarantees end and an open question begins.

Appendix A: On-Chain Reference Addresses

The following on-chain addresses are referenced throughout this paper by description rather than repeated inline; they are collected here for readers who wish to verify any claim directly against the Solana ledger.

Reference	Address / Identifier
ACT mint address	BmaBEY6NDbLevUcU59Fgie8JHeqD4UjEFJda8LuSS2yT
Token-2022 program	TokenzQdBNbLqP5VEhdkAS6EPFLC1PHnBqCXEPXuEb
Treasury multisig (Squads)	3FyoJdvc7FaZDEt5YoLHF3PB2xo3vtp4GWTBTN6cyzZg
Management fee wallet	2tcBrd1JQjL8VHNFRYB1EurbyLiVAKZTYTYk94aVoZX2

The comparative trading-volume and market-capitalization figures in footnotes 5 through 7 are drawn from third-party market-data aggregators currently accessible at the time of writing, are subject to constant change, and have not been independently re-verified against primary on-chain data by this paper's authors. They are included to illustrate the general scale and cross-chain distribution of permissionless trading activity, not as precise or current figures to be relied upon.

This whitepaper is preliminary and subject to change. It does not constitute an offer or solicitation to buy or sell any security, token, or other financial instrument in any jurisdiction where such offer or solicitation would be unlawful. This protocol has not been registered under the securities laws of any jurisdiction. Prospective holders should conduct their own research and consult independent advisors before making any decision related to ACT.